An L^1 counterexample at critical logarithmic BSDE growth

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Abstract

For the generator $g(z) = |z|/\sqrt{\log(e+|z|)}$, we construct a nonnegative terminal value of expectation one for which the scalar BSDE has no class-(D) solution. The construction works on every finite positive horizon and in every Brownian dimension. It gives a negative answer to Fan, Hu and Tang's Problem 5.1.

Let B be a d-dimensional Brownian motion, $d \geq 1$, on the Brownian basis of [1], and fix $T > 0$. Class (D) means that $\{Y_\tau : \tau \leq T \text{ a stopping time}\}$ is uniformly integrable. A solution has continuous adapted Y, predictable Z, and $\int_0^T |Z_t|^2 dt < \infty$ almost surely, with the BSDE identity below. All logarithms are natural; vector norms are Euclidean.

Theorem 1. *There is an explicitly specified $\mathcal{F}_T$-measurable $\xi \geq 0$ with $\mathbb{E}\xi = 1$ for which no class-(D) solution satisfies*

$$Y_t = \xi + \int_t^T g(Z_s)\, ds - \int_t^T Z_s \cdot dB_s, \qquad g(z) = \frac{|z|}{\sqrt{\log(e+|z|)}}. \tag{1}$$

We first record why bounded lower solutions can be compared with a putative class-(D) solution despite a change of measure.

Lemma 1. *If (Y, Z) solves (1) with $\xi \geq 0$ and Y of class (D), then $Y \geq 0$. Suppose a continuous adapted process u satisfies $0 \leq u \leq C$ for a deterministic finite constant C, and*

$$du_t = -f_t\, dt + z_t \cdot dB_t, \quad \int_0^T |z_t|^2 dt < \infty, \quad f_t \leq g(z_t), \quad u_T \leq \xi.$$

Then $u \leq Y$.

Proof. Since $g \geq 0$, localization of (1) gives $Y_t \geq \mathbb{E}[Y_{\tau_n} \mid \mathcal{F}_t]$ for stopping times $\tau_n \uparrow T$. The stopped stochastic integrals may be chosen square-integrable. Class (D) and continuity give $Y_{\tau_n} \to \xi$ in L^1, whence $Y_t \geq \mathbb{E}[\xi \mid \mathcal{F}_t] \geq 0$. Taking rational t and using continuity makes this assertion simultaneous in time.

For $h(r) = r/\sqrt{\log(e+r)}$, $0 \leq h'(r) \leq 1$ on $r > 0$. Thus g is globally 1-Lipschitz, also in $\mathbb{R}^d$. Put $\Delta = u - Y$ and choose the predictable vector

$$\theta_t = \begin{cases} \dfrac{g(z_t) - g(Z_t)}{|z_t - Z_t|^2}(z_t - Z_t), & z_t \neq Z_t, \\ 0, & z_t = Z_t. \end{cases}$$

Then $|\theta_t| \leq 1$. Novikov's condition holds, so under the equivalent measure with density $\exp(\int_0^T \theta_t \cdot dB_t - \frac{1}{2}\int_0^T |\theta_t|^2 dt)$, the process $B^\theta = B - \int_0 \theta_t dt$ is Brownian. Writing $r_t = g(z_t) - f_t \geq 0$, we obtain

$$d\Delta_t = r_t dt + (z_t - Z_t) \cdot dB_t^\theta.$$

Tanaka's formula makes Δ^+ a local submartingale. It is bounded by C, because $Y \geq 0$, hence is a true submartingale. Its terminal value is zero, so it vanishes identically. This argument needs no integrability of Y or ξ under the changed measure. □

Bounded Gaussian lower solutions

Write Φ and ϕ for the standard normal distribution function and density. Set

$$S = \min\{T, (2\pi)^{-1}\}, \qquad H_N = \sum_{k=1}^{N} \frac{1}{k}, \qquad \delta_k = \frac{S}{kH_N}, \qquad t_k = \sum_{j=1}^{k} \delta_j.$$

For each integer $N \geq 1$, define

$$X_N = 2^N \prod_{k=1}^{N} \Phi\left(\frac{B^1_{t_k} - B^1_{t_{k-1}}}{\sqrt{\delta_k}}\right). \tag{2}$$

The standardized increments are independent standard normals, and $\mathbb{E}\Phi(G) = 1/2$ by symmetry. Consequently $\mathbb{E}X_N = 1$.

Define deterministic constants backwards, starting with $k = N$:

$$D_k = (2\pi\delta_k)^{-1/2}, \quad L_k = 2^N D_k \prod_{j>k} p_j, \quad q_k = [\log(e + L_k)]^{-1/2}, \quad p_k = \Phi(q_k\sqrt{\delta_k}/\sqrt{2}). \tag{3}$$

This recursion uses only constants already defined. For $t_{k-1} \leq t \leq t_k$, put

$$A_k = 2^N \prod_{j<k} \Phi(\Delta B^1_j/\sqrt{\delta_j}) \prod_{j>k} p_j, \qquad w_t = \frac{B^1_t - B^1_{t_{k-1}} + q_k(t_k - t)}{\sqrt{\delta_k + t_k - t}},$$

where $\Delta B^1_j = B^1_{t_j} - B^1_{t_{j-1}}$, and define

$$u^N_t = A_k\Phi(w_t), \qquad z^N_t = \frac{A_k\phi(w_t)}{\sqrt{\delta_k + t_k - t}}\, e_1.$$

At t_k, the next stage starts with its Gaussian factor equal to p_{k+1}, so the values of u^N agree. After S, set $u^N = X_N$ and $z^N = 0$. Direct Itô differentiation on each interval gives

$$du^N_t = -q_k|z^N_t|dt + z^N_t \cdot dB_t, \qquad 0 \leq |z^N_t| \leq L_k. \tag{4}$$

The finite collection of deterministic bounds gives square-integrability of z^N. Moreover $0 \leq u^N \leq 2^N$, $u^N_T = X_N$, and

$$q_k|z^N_t| \leq g(z^N_t), \qquad u^N_0 = \prod_{k=1}^{N} 2p_k.$$

Thus u^N satisfies the bounded lower-solution conditions of Lemma 1.

Divergence of the initial values

Put $x_k = q_k\sqrt{\delta_k}/\sqrt{2}$ and $\ell_k = \log(2p_k)$. Since $D_k \geq 1$ and $p_j \geq 1/2$, we have $L_k \geq 2^k$ and $q_k \leq (k\log 2)^{-1/2}$. Also $q_k \leq 1$, so $0 \leq x_k \leq 1$. On this interval,

$$\phi(1)x \leq \log(2\Phi(x)) \leq \sqrt{2/\pi}\, x. \tag{5}$$

Indeed the derivative is $\phi(x)/\Phi(x)$, between $\phi(1)$ and $2\phi(0)$. It follows that

$$\sum_{j>k} \ell_j \leq \sqrt{\frac{S}{\pi\log 2}}\,\sqrt{H_N}. \tag{6}$$

Set $a = \log(1/(2\pi S)) \geq 0$. For $k \geq \lceil H_N \rceil$ and $H_N \geq \max\{8, a\}$, use

$$\log L_k = k \log 2 + \tfrac{1}{2} \log(kH_N) + \tfrac{1}{2} a + \sum_{j>k} \ell_j.$$

The coefficient in (6) is at most one. Since $L_k \geq 2^k$, $\log(e + L_k) \leq 1 + \log L_k$, and hence

$$\log(e + L_k) \leq 1 + k \log 2 + \log k + k/2 + \sqrt{k} \leq 5k.$$

The lower bound in (5) now gives

$$\begin{aligned} \log u_0^N &\geq \phi(1) \sqrt{\frac{S}{10 H_N}} \sum_{k=\lceil H_N \rceil}^{N} \frac{1}{k} \\ &\geq \frac{\phi(1)}{2\sqrt{10}} \sqrt{S H_N}. \end{aligned} \tag{7}$$

For the last inequality, the omitted harmonic sum is at most $1 + \log H_N \leq H_N/2$ when $H_N \geq 8$.

One integrable terminal value

For $m \geq 1$, choose the explicit integers

$$N_m = \left\lceil \exp\left(8 + a + \frac{160 m^2 (\log 2)^2}{S \phi(1)^2} \right) \right\rceil, \qquad \xi = \sum_{m=1}^{\infty} 2^{-m} X_{N_m}.$$

Tonelli's theorem and (2) give $\mathbb{E}\xi = 1$, so this sum is finite almost surely. The summands need not be independent. Since $H_N \geq \log(N+1)$, (7) gives $u_0^{N_m} \geq 2^{2m}$.

Suppose a class-(D) solution with terminal value ξ existed. For $0 < c \leq 1$, $g(cz) \geq c g(z)$, because the logarithmic denominator decreases under scaling. Hence $2^{-m} u^{N_m}$ is still a bounded nonnegative lower solution, with terminal value at most ξ. Lemma 1 yields

$$Y_0 \geq 2^{-m} u_0^{N_m} \geq 2^m \qquad (m \geq 1).$$

These countably many inequalities contradict the finiteness of Y_0. This proves Theorem 1 and the negative answer to [1, Problem 5.1, equation (5.1)].

Scope. The generator, exponent, Brownian basis and class-(D) condition are unchanged. Only an admissible terminal value is chosen to refute the universal existence claim.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

References

[1] S. Fan, Y. Hu and S. Tang. *1D nonlinear backward stochastic differential equations: a unified theory and applications.* arXiv:2309.06233v2, 11 February 2026, §1 and §5, Problem 5.1. Primary text.